Category 1: Project Scope & Planning (IS 875 & NBC 2016)

1. What is the primary engineering objective of selecting the "Indiramma Illu" scheme for your mini-project?
2. You mentioned a budget constraint of ₹5,00,000. As a civil engineer, how did you ensure your structural design did not inflate this cost?
3. According to NBC 2016, under which exact group and subdivision does this building fall?
4. Your gross plot area is 79.73 sq.m, but your net plot area is 54.82 sq.m. Explain the specific setback rules from G.O.Ms.No.168 you applied to arrive at this.
5. Why did you provide a 1.5m setback at the front but only 0.5m on the sides?
6. Explain the difference between Plinth Area and Carpet Area. What is the mandatory minimum built-up area for this government scheme?
7. You restricted the overall depth of the slab to 100mm. Is this the minimum prescribed by IS 456? Justify your choice structurally.
8. What is the minimum headroom clearance you maintained under the structural beams, and what does the code dictate?
9. You planned for a road widening width of 4.5m from the center line. What happens to the structural footprint if the municipality decides to widen the road tomorrow?
10. Why did you opt for load-bearing masonry for the walls instead of a fully framed RCC structure for a G+0 building?

Category 2: Load Calculations & Distribution

11. You assumed the unit weight of RCC as 25 KN/m^3 and brick masonry as 18 KN/m^3. Which specific part of IS 875 gives you these values?
12. For the roof slab, you took the live load as 1.5 KN/m^2, but for the floor slab, you took 2.0 KN/m^2. Explain this discrepancy.
13. You stated you ignored Earthquake loads because Mahabubnagar is not in an earthquake-prone area. According to IS 1893, which seismic zone does Mahabubnagar fall under? Should it be entirely ignored?
14. Explain the criteria for load distribution from a two-way slab to the supporting beams. Why does it form triangles and trapezoids?
15. What are the specific formulas you used to calculate the equivalent Uniformly Distributed Load (UDL) on beams from a trapezoidal load distribution?
16. How did you calculate the weight of the weathering course, and why is it factored into the dead load of the roof slab?

Category 3: Structural Analysis (Kani's Method)

17. You chose Kani's Method over the Moment Distribution Method. What is the primary mathematical advantage of Kani's Method for multi-storey frames?
18. Define Rotation Factor. How did you calculate the sum of stiffness (∑ k) at joint B1?
19. When calculating stiffness (k), you used 4EI/L. Under what support condition is this valid?
20. In your Frame A1-B1-C1 analysis, you mentioned fixed end moments in all columns are zero. Why?
21. What is a Displacement Factor, and why did you get a value of -0.5 for your columns in Storey 1?
22. Explain the process of finding the "Joint Rotation" contribution in any given cycle of Kani's method.
23. In Cycle 4 for Frame A1-B1-C1, how did you verify that the joint was in equilibrium to halt the iterations?
24. How did you calculate the Final End Moments from the Fixed End Moments, Near End Rotation, and Far End Rotation?
25. Looking at your Bending Moment diagram for Frame A1-B1-C1, why is the moment at support B1 significantly higher (22.30 KN-m) than at A1?
26. If you had to analyze this same frame using STAAD.Pro, what differences would you expect in your final bending moment values, and why?

Category 4: Limit State Method & Slab Design (IS 456:2000)

27. Why did you use the Limit State Method instead of the Working Stress Method for this project? What is the specific drawback of WSM?
28. What are the partial safety factors for material strength of concrete and steel in the Limit State of Collapse?
29. You chose M20 concrete and Fe500 steel. What does the "500" in Fe500 stand for, and what is its limiting depth of neutral axis ($x_{u, max}$)?
30. For Slab S1, your Ly/Lx ratio is 1.19. What dictates that this must be designed as a two-way slab?
31. How did you determine the effective depth of the slab along the shorter span versus the longer span? Why are they different?
32. You calculated the factored bending moments using αx and αy coefficients. From which table in IS 456 did you extract these, and what boundary conditions do they represent?
33. Your calculated required depth for the slab was 32.34mm, but you provided 100mm. Why the massive over-design?
34. For the deflection check of the two-way slab, you used a Basic Value (BV) of 40. Where does this value come from?

35. How did you calculate the Modification Factor (MF) for tension reinforcement in your deflection check?

Category 5: Beam Design & Shear Reinforcement

36. In your continuous beam design, how did you ensure that Mu < Mu, limit}? What would you have to do if Mu exceeded Mu, limit?
37. Explain the formula used to calculate the area of tension steel (Ast) for a singly reinforced section.
38. For span A1-B1, you checked for shear. How is the nominal shear stress (τv) calculated?
39. Your τv exceeded τc at joint B1. What is τc dependent on, and how did you extract it from Table 19 of IS 456?
40. You provided 8mm dia 2-legged vertical stirrups at 275mm c/c. State the three IS code limit conditions for the maximum spacing of shear reinforcement.
41. What is the significance of τc, max? What would you do if your nominal shear stress exceeded τc, max?

Category 6: Column Design (Uni-axial & Bi-axial Bending)

42. You designed Column A1 for bi-axial bending. Define bi-axial bending and explain how it differs from uni-axial bending in Column B1.
43. You assumed a reinforcement percentage (p) of 1.17% for Column A1. What is the minimum and maximum percentage of steel allowed in a column?
44. Explain the interaction equation (Bresler's empirical formulation) used to check the safety of a bi-axially loaded column.
45. In utilizing SP-16 Chart 48, you used the parameters Pu / fck . bD and Mu fckbD^2. What exactly do these non-dimensional parameters represent?
46. How did you determine the pitch (spacing) of the lateral ties for the columns? What three criteria did you check?

Category 7: Footing & Lintel Design

47. For Footing A1, your Safe Bearing Capacity (SBC) is 200 KN/m^2. How does SBC impact the size of your footing?
48. Explain the exact location of the critical section for **One-Way Shear** in an isolated footing.
49. Explain the exact location of the critical section for **Two-Way (Punching) Shear** in an isolated footing.
50. In your lintel design, you assumed a 60° load dispersion angle. Why didn't you consider the full rectangular load of the masonry wall above the opening?